MRC Lecture 1 — Derivation Guide 1

The Dot Product: Algebraic and Geometric Definitions

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1. Motivation

Every compatibility score in this course — between an antibody and an antigen, between a transcription factor and a DNA site, between a transformer query and a key — is computed as a **dot product** of two vectors. The dot product is the atomic operation of molecular recognition computing. This guide derives the dot product from both its algebraic and geometric definitions, proves their equivalence, and works through examples with molecular interpretation.

2. The Algebraic Definition

Definition. Given two vectors $a=(a_1, a_2, \ldots, a_n)$ and $b=(b_1, b_2, \ldots, b_n)$ in $\mathbb{R}^n$, their dot product (also called inner product or scalar product) is:

$$a \cdot b=\sum_{i=1}^{n} a_i b_i=a_1 b_1+a_2 b_2+\cdots+a_n b_n$$

The dot product takes two vectors as input and produces a single real number (a scalar) as output.

Properties of the dot product:

(i) Commutativity: $a \cdot b=b \cdot a$
(ii) Distributivity: $a \cdot (b+c)=a \cdot b+a \cdot c$
(iii) Scalar multiplication: $(\alpha a) \cdot b=\alpha (a \cdot b)$
(iv) Positive definiteness: $a \cdot a \geq 0$, with equality if and only if $a=0$

Property (iii) is called **bilinearity** (linearity in both arguments). This property will become central in Lecture 5, where we show that the binding energy between two molecules decomposes as a bilinear function of their features — which is precisely a dot product.

3. Worked Example 1: Numerical Computation

Problem. Compute $a \cdot b$ for $a=(3,1,2)$ and $b=(1,4,-1)$.
Step 1. Multiply corresponding components:

$$a_1 b_1=3 \times 1=3$$
$$a_2 b_2=1 \times 4=4$$
$$a_3 b_3=2 \times (-1)=-2$$

Step 2. Sum the products:

$$a \cdot b=3+4+(-2)=5$$

The result is a single number: 5. The vectors are “mostly aligned” — their dot product is positive.

4. The Geometric Definition

Definition. The dot product can equivalently be written as:

$$a \cdot b=\| a \| \| b \| \cos\theta$$

where $\| a \|$ is the length (Euclidean norm) of a, $\| b \|$ is the length of b, and θ is the angle between the two vectors.
The Euclidean norm. The length of a vector $a = (a_1, a_2, \ldots, a_n)$ is:

$$\| a \| = \sqrt{\sum_{i=1}^{n} a_i^2} = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$$

Note that $\| a \|^2 = a \cdot a$, i.e., the squared length of a vector is its dot product with itself.

5. Proof of Equivalence

We prove that the algebraic and geometric definitions are consistent. The key tool is the law of cosines.
Setup. Consider vectors a and b in $\mathbb{R}^n$ with angle θ between them. The vector from the tip of b to the tip of a is $a - b$.
Step 1. The law of cosines applied to the triangle formed by a, b, and $a - b$ gives:

$$\| a - b \|^2 = \| a \|^2 + \| b \|^2 - 2 \| a \| \| b \| \cos\theta$$

Step 2. Expand the left side using the algebraic definition:

$$\begin{aligned} \| a - b \|^2 &= (a - b) \cdot (a - b) \\ &= a \cdot a - 2(a \cdot b) + b \cdot b \\ &= \| a \|^2 - 2(a \cdot b) + \| b \|^2 \end{aligned}$$

Step 3. Set the two expressions equal:

$$\| a \|^2 - 2(a \cdot b) + \| b \|^2 = \| a \|^2 + \| b \|^2 - 2 \| a \| \| b \| \cos\theta$$

Step 4. Cancel $\| a \|^2 + \| b \|^2$ from both sides:

$$-2(a \cdot b) = -2 \| a \| \| b \| \cos\theta$$

Step 5. Divide by -2:

$$a \cdot b = \| a \| \| b \| \cos\theta \quad \blacksquare$$

The two definitions are equivalent. The algebraic definition is what you compute. The geometric definition is what it means.

6. Geometric Interpretation

The geometric definition reveals three regimes:
Case 1: $\theta = 0°$ (parallel vectors).

$$a \cdot b = \| a \| \| b \| \cos 0° = \| a \| \| b \| \times 1 = \| a \| \| b \|$$

The dot product is maximized. The vectors point in the same direction. In a molecular embedding space, this means the two molecules are maximally compatible — they are predicted to bind.
Case 2: $\theta = 90°$ (orthogonal vectors).

$$a \cdot b = \| a \| \| b \| \cos 90° = \| a \| \| b \| \times 0 = 0$$

The dot product is zero. The vectors are perpendicular. The two molecules share no relevant features — they are unrelated.
Case 3: $\theta = 180°$ (antiparallel vectors).

$$a \cdot b = \| a \| \| b \| \cos 180° = \| a \| \| b \| \times (-1) = -\| a \| \| b \|$$

The dot product is minimized (most negative). The vectors point in opposite directions. The two molecules are maximally incompatible.
Summary: The dot product measures alignment. Positive means compatible. Zero means unrelated. Negative means incompatible. This interpretation carries through the entire course.

7. Worked Example 2: Geometric Computation

Problem. For $a = (3, 1, 2)$ and $b = (1, 4, -1)$ (same vectors as Example 1), verify the geometric formula and find the angle between them.
Step 1. Compute the norms:

$$\|a\| = \sqrt{3^2 + 1^2 + 2^2} = \sqrt{9 + 1 + 4} = \sqrt{14} \approx 3.742$$
$$\|b\| = \sqrt{1^2 + 4^2 + (-1)^2} = \sqrt{1 + 16 + 1} = \sqrt{18} \approx 4.243$$

Step 2. Use the algebraic result from Example 1: $a \cdot b = 5$.
Step 3. Solve for $\cos\theta$:

$$\cos\theta = \frac{a \cdot b}{\|a\| \|b\|} = \frac{5}{\sqrt{14} \times \sqrt{18}} = \frac{5}{\sqrt{252}} = \frac{5}{15.875} \approx 0.315$$

Step 4. Find the angle:

$$\theta = \arccos(0.315) \approx 71.6°$$

The vectors are at about 72° — partially aligned but far from parallel. The dot product (5) is positive, consistent with this angle being less than 90°.

8. Worked Example 3: Molecular Interpretation

Problem. Consider a simplified 3-dimensional binding feature space where the three dimensions represent: (1) electrostatic complementarity, (2) shape complementarity, and (3) hydrophobic contact strength. An antibody has feature vector $a_{\text{Ab}} = (2, -1, 3)$. Two candidate antigens have feature vectors $b_{\text{Ag1}} = (1, -2, 2)$ and $b_{\text{Ag2}} = (-1, 3, -1)$. Which antigen is the antibody more likely to bind?
Step 1. Compute the dot product with Antigen 1:

$$a_{\text{Ab}} \cdot b_{\text{Ag1}} = 2(1) + (-1)(-2) + 3(2) = 2 + 2 + 6 = 10$$

Step 2. Compute the dot product with Antigen 2:

$$a_{\text{Ab}} \cdot b_{\text{Ag2}} = 2(-1) + (-1)(3) + 3(-1) = -2 - 3 - 3 = -8$$

Step 3. Interpret:
The dot product with Antigen 1 is $+10$ (strongly positive) — the antibody and Antigen 1 have aligned features across all three dimensions. Their electrostatic properties are complementary (both positive), their shape features are complementary (both negative, indicating matching concavity), and their hydrophobic contacts are aligned (both positive and large).
The dot product with Antigen 2 is -8 (negative) — the antibody and Antigen 2 have opposing features. The electrostatic term is unfavorable (opposite signs), shape complementarity is poor, and hydrophobic contacts are mismatched.
Conclusion: The antibody is predicted to bind Antigen 1 (score $= +10$) and not Antigen 2 (score $= -8$). When we reach Lecture 6, these scores will be fed into the softmax function (the convergence equation) to produce binding probabilities.

9. The Dot Product as Projection

There is a third way to interpret the dot product that is useful for understanding attention mechanisms. The **scalar projection** of a onto b is:

$$\text{proj}_b(a) = \frac{a \cdot b}{\|b\|}$$

This gives the signed length of the shadow that a casts onto b. The dot product itself is this projection scaled by $\|b\|$:

$$a \cdot b = \text{proj}_b(a) \times \|b\| = \|a\| \cos\theta \times \|b\|$$

Why this matters for attention. In transformer attention, the query vector q is projected onto each key vector k_j via the dot product. The projection measures how much the query "overlaps" with each key. The key with the largest projection receives the most attention. This is exactly the same operation as asking: which target's features overlap most with the agent's features? The answer determines binding probability.

10. Summary

Concept	Definition	Interpretation
Algebraic dot product	$a \cdot b = \sum_i a_i b_i$	Sum of component-wise products
Geometric dot product	$a \cdot b = \|a\| \|b\| \cos\theta$	Product of lengths times cosine of angle
Positive dot product	$a \cdot b > 0$	Vectors are aligned ($\theta < 90°$); molecules are compatible
Zero dot product	$a \cdot b = 0$	Vectors are orthogonal ($\theta = 90°$); molecules are unrelated
Negative dot product	$a \cdot b < 0$	Vectors are opposed ($\theta > 90°$); molecules are incompatible
Bilinearity	$(\alpha a) \cdot b = \alpha (a \cdot b)$	Energy decomposes linearly in both partners (Lecture 5)
Projection	$\text{proj}_b(a) = (a \cdot b) / \|b\|$	How much a overlaps with b (attention mechanism)

11. Checkpoint Questions

Q1. Compute $a \cdot b$ for $a = (1, -2, 3, 0)$ and $b = (4, 1, -1, 2)$.
Answer: $1(4) + (-2)(1) + 3(-1) + 0(2) = 4 - 2 - 3 + 0 = -1$. The vectors are slightly opposed.
Q2. Two vectors have norms $\|a\| = 5$ and $\|b\| = 3$ and their dot product is $a \cdot b = 15$. What is the angle between them?
Answer: $\cos\theta = 15 / (5 \times 3) = 1.0$, so $\theta = 0°$. The vectors are perfectly parallel.
Q3. An antibody embedding has dot product $+12$ with Antigen A and dot product $+3$ with Antigen B. Both dot products are positive. Does this mean the antibody binds both antigens?
Answer: Both scores are positive, so the antibody has some compatibility with both antigens. But the relevant question is not the absolute score — it is the relative score. When these scores are passed through softmax (Lecture 6), the probability of binding Antigen A will be much higher than binding Antigen B because softmax exponentiates the scores before normalizing, amplifying the difference. The dot product gives the raw compatibility; softmax converts it to a selection probability.
Q4. Why is the bilinearity property of the dot product important for molecular recognition?
Answer: Bilinearity means that the dot product is linear in both arguments: $(\alpha a + \beta c) \cdot b = \alpha (a \cdot b) + \beta (c \cdot b)$. In molecular binding, this corresponds to the energy decomposing as a sum of independent contributions from each contact position — the energy at position p is a product of the agent feature at p and the target feature at p, and the total energy is the sum over all positions. This bilinear decomposition is what bridges the Boltzmann distribution to the transformer dot-product attention equation (Lectures 5 and 7).

12. Connection to Future Lectures

The dot product introduced here appears in every subsequent lecture:

- **Lecture 2:** The transformer attention score $q_i \cdot k_j$ is a dot product between learned query and key vectors.
- **Lecture 3:** The InfoNCE contrastive loss uses $\text{sim}(z_i, z_j) = z_i \cdot z_j$ (after L2 normalization) as the similarity function.
- **Lecture 4:** The Boltzmann binding energy $E_j = -q \cdot k_j$ identifies the dot product as the dimensionless binding energy.

- **Lecture 5:** The position weight matrix for TF–DNA binding is a dot product in raw sequence space. The SantaLucia nearest-neighbor model for RNA/DNA is a sum of positional energy terms — a generalized dot product.
- **Lecture 7:** The five architecture conditions derive the SFM architecture. Condition 2 (bilinear energy decomposition) is the condition that makes the dot product the correct compatibility measure.